

**REGULARIZING CONTROL INVARIANCE:
CONTROL BARRIER FUNCTIONS AND WHERE TO FIND THEM**

by

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ABSTRACT

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Regularizing Control Invariance: Control Barrier Functions and Where to Find Them

A dissertation directed by Associate Professor Marco M. Nicotra

The control invariance property of sets was characterized in terms of the Bouligand contingent cone by Nagumo in 1942. Under standard regularity and transversality conditions, the expression for the contingent cone simplifies significantly at the boundary. This dissertation demonstrates how control barrier functions extend this characterization to the entire set beyond its boundary: a process termed regularization. In doing so, the dissertation exposes the geometric conditions under which control barrier functions exist. These existence conditions serve as the safety equivalent to the converse Lyapunov theorems for stability. Finally, to bridge theory and application, the dissertation highlights how tools from the reference governor literature can be used to obtain control barrier functions.

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FOREWORD

At the time of writing this dissertation, the rapid development of artificial intelligence (AI) is changing our relationship with autonomous technology. We are entering an era where the volume of AI-driven decisions often outpaces our ability to verify their safety. While modern AI can make highly optimized decisions, it often operates as a closed box lacking formal guarantees. When human lives or critical infrastructure depend on these systems, empirical trust is insufficient.

The constrained control field offers a blueprint for how we might begin to address this. Through mathematical properties like control invariance and constructs like control barrier functions (CBFs), we can establish certificates of safety: proof that a dynamical system will never violate its imposed boundaries, regardless of the underlying complexity.

The research in this dissertation explores the connections between control invariance and CBFs. Moreover, it leverages tools from the reference governor literature to find CBFs. While artificial intelligence is not explicitly discussed in the chapters that follow, the theoretical foundations developed herein provide the rigorous mathematical groundwork necessary to one day guarantee the safety of AI-driven autonomous systems.



*Original artwork by Michele Carminati,
commissioned by Marco M. Nicotra.*

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The content of this dissertation is based on the published works [1–7].

The complete dissertation will be available once all associated research is published.

Part I

Continuous Time

Part II

Discrete Time

Part III

Appendices

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